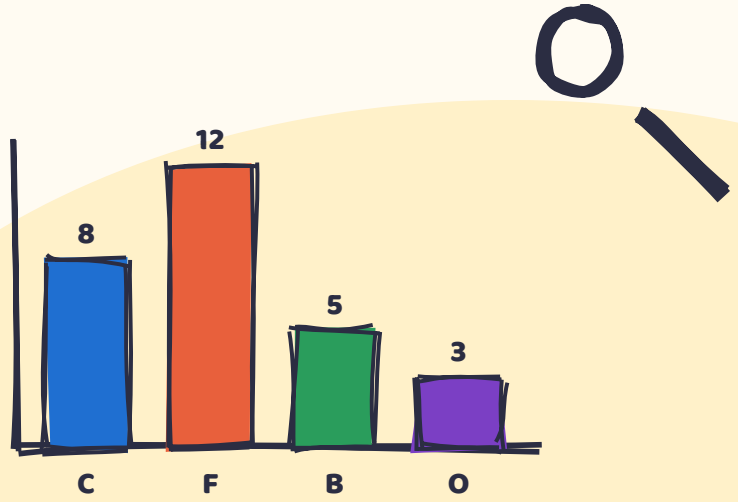
Junior Makers · Class 30

Data Detectives
Class Survey Charts

Ask a question, collect clues, solve it with charts (C = cricket, F = football…)

Fair questions | Tally & tables | CODAP charts | Evidence



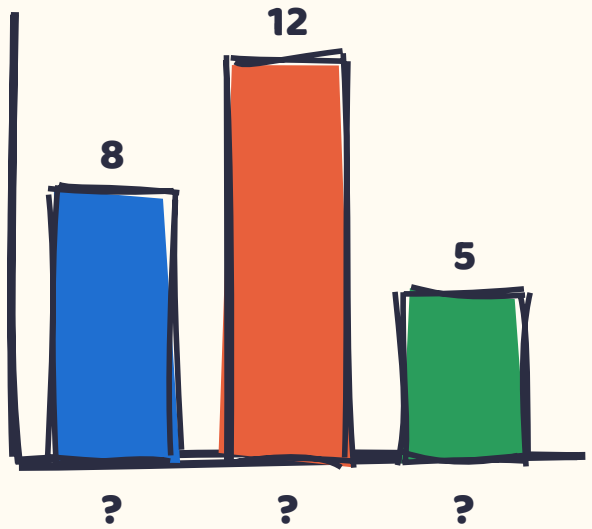
Did you know?
Doctors used a simple chart in 1854 to discover that dirty water was spreading cholera in London. Data saves lives!

Let's think...

What's our class's favourite sport?

Everyone has a guess

Ask five people and you'll get five different answers: 'Everyone loves cricket!' 'No, football!' How can we find out for **sure**, and convince someone who disagrees? Detectives use **evidence**.



Shout loudest?

Ask one friend?

Collect data!

Today: you design a **survey**, collect your class's answers, and solve your question with **charts** in CODAP.

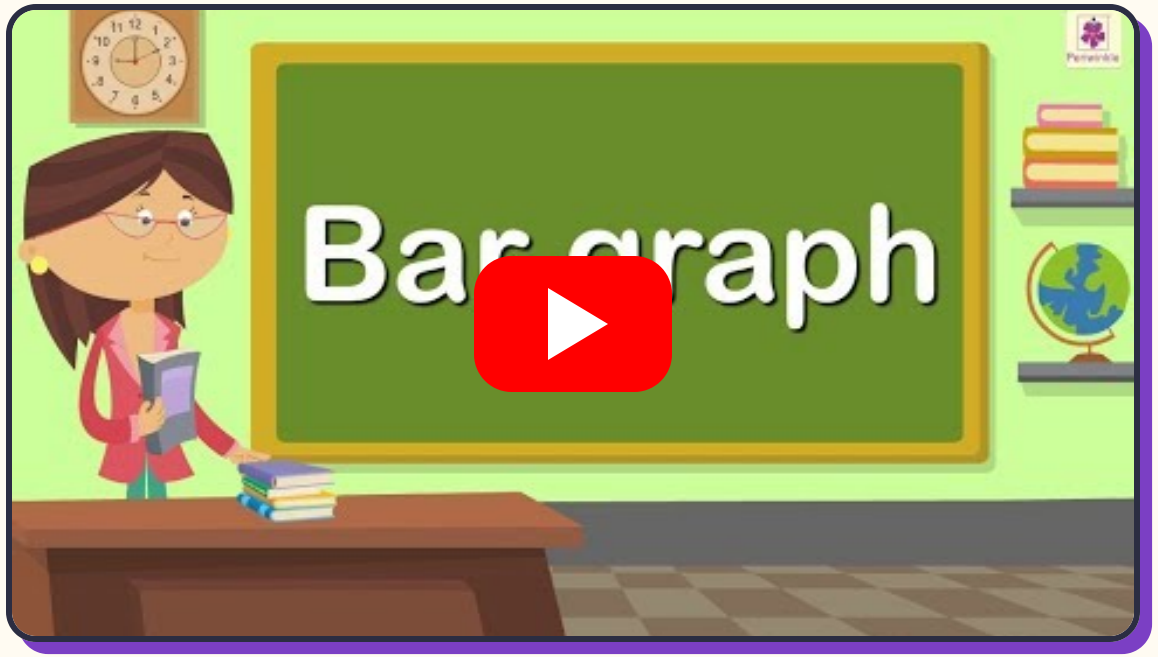
Learning objectives

By the end of this class I can...

1	**Good questions:** write a **fair survey question** with clear answer choices	**Proof:** our question passed the 'fair question' checklist
2	**Collect data:** collect data with a **tally** and a **table**	**Proof:** our table has every classmate's answer
3	**Charts:** make **bar charts** and **dot plots** in CODAP and read them	**Proof:** our chart answers our question
4	**Conclude:** make a **claim with evidence** and spot a misleading chart	**Proof:** one sentence: 'Most... because the chart shows...'

Objective 3 · Charts · video

Video time!



Bar Graph

Periwinkle · about 4 minutes · youtu.be/J2DKgCf353k

Watch for...

1. What do the **axes** of a bar graph show?
2. How do you find the **biggest** and **smallest** value?
3. Why is a graph easier to read than a list of numbers?

Scan to watch

After the video: Think of one question about our class you'd like answered with a graph.

Objective 1 · Good questions

Ask a fair question

The idea

A good survey question:

- is **clear**: everyone understands it the same way
- has **answer choices** that cover everyone (add 'other')
- is **fair**: it doesn't push people towards one answer

❌ 'Don't you agree cricket is the best sport?'
✅ 'Which sport do you most like to play? cricket / football / badminton / other'

Example: a fair referee

✗ leading question

✓ fair, with choices

A good referee doesn't cheer for one team. A good question doesn't 'cheer' for one answer either. 'Don't you agree...' is like a referee wearing one team's shirt!

LEARN

Objective 2 · Collect data

Collect: tally, then table

The idea

Ask everyone and make a **tally**: one line per answer, and every 5th line crosses the group (𝍸). Groups of 5 are fast to count!

Then put it in a **table**: one row per answer choice, with the total. Check: does the total equal the number of people asked?

Example: counting votes in an election



football: 12



cricket: 8

Election counters make tally marks in groups of 5 so they don't lose count. Then the totals go in a table that everyone can check. That's exactly what we're doing.

Objective 3 · Charts

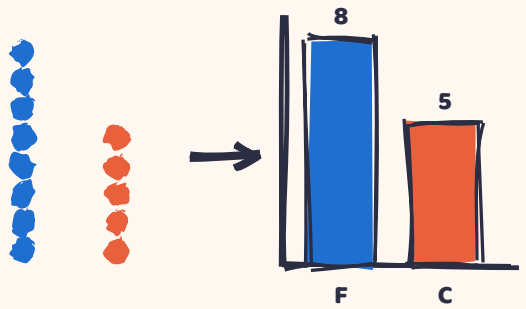
Dot plots and bar charts

The idea

In CODAP, each **dot** is one real person. Stack them and you get a **dot plot**. Click the ruler menu → **Fuse dots into bars** and it becomes a **bar chart**.

- **Tallest bar** = most common answer (the **mode**)
- Compare bars: 'football has 4 more than cricket'
- Click a dot to see which row it is!

Example: a queue at the canteen



If you lined up by favourite snack, you could SEE which queue is longest. A dot plot is those queues lying down. A bar chart draws a box round each queue.

Objective 4 · Conclude

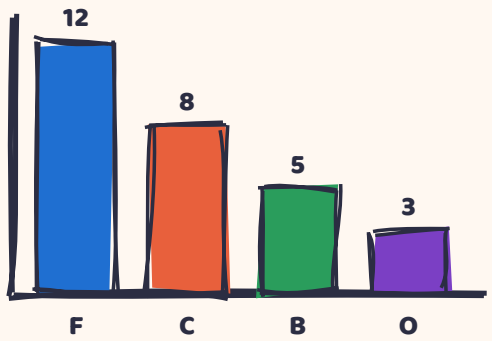
Claim + evidence = a detective's answer

The idea

A good answer has a **claim** and **evidence** from the chart:

Claim: Football is our class's favourite sport.
Evidence: 12 out of 28 students chose it, more than any other sport. Cricket was second with 8.
But: we only asked one class, so we can't say it's true for the whole school.

Example: a detective in court

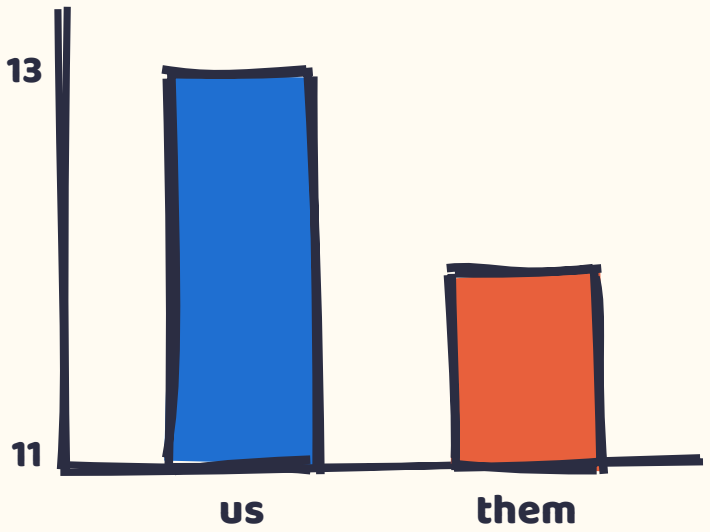


A detective can't just say 'the butler did it!'. They must show clues. Your chart is your clue, and the numbers are your proof.

LEARN

Objective 4 · Conclude · spot the trick

Charts can mislead



What's the trick?

The side axis starts at **11**, not 0! 13 vs 12 looks like 'us' is twice as big, but it's only 1 more!

Detective checklist

- ✔ Does the axis start at 0?
- ✔ How many people were asked?
- ✔ Is the question fair?

Real world

Who uses data to decide?



Doctors
Find which medicines work best



Weather forecasters
Spot patterns in years of weather



Cricket teams
Choose players from batting and bowling stats



Farmers
Pick crops from rainfall and soil data

Pakistan's cricket team uses data analysts. What data would help choose a bowler?

AI needs data
AI learns from data too. Bad data teaches AI wrong things, just like a bad survey.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Fix the question
2. Plan your investigation
3. Survey your class
4. Put your data in CODAP
5. Compare two groups
6. Detective report
7. Detective's checklist

Class game

Fix the question

How to play

1. Teacher shows 6 bad survey questions, one at a time.
2. Teams have 30 seconds to say **what's wrong** and rewrite it.
3. e.g. 'How many hours do you play games? 0-1, 1-2, 2-3' (where does 1 go? and what about 5 hours?)
4. e.g. 'Is homework bad?' (leading, and only yes/no)
5. Best rewrite wins a point.

Then discuss

Which mistake was most common?

Why does it matter for our results?



Big idea

Bad questions give bad data. Bad data gives wrong answers, however nice the chart looks.

Objective 1 · Good questions · plan

Plan your investigation

Our plan	
Our question	
Answer choices	
What we predict	
Who we'll ask	everyone in our class
One extra fact to collect	e.g. age or boy/girl, to compare groups

Ideas

Favourite fruit · way to school · hours of sleep · favourite subject · number of siblings · bedtime

MAKE

Objective 2 · Collect data · collect

Survey your class

Answer	Tally	Total
Total		

Rules

Ask **everyone**, once each. Write exactly what they say. Don't argue with an answer!

Check!

Does your total = number of people in class? If not, someone was missed or counted twice.

MAKE

Objective 3 · Build step 1 of 2

Put your data in CODAP

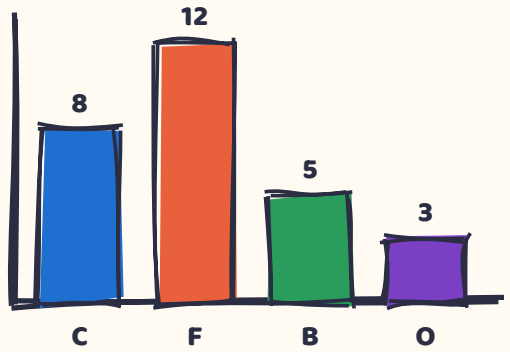
a Open **codap.concord.org** → Create New Document. (No account needed.)

b Click **Tables** → New. Name the columns, e.g. **name** and **sport**.

c Type one row per person. (Or use first names only / initials for privacy.)

d Click **Graph**. A blank graph appears.

e Drag the **sport** column title onto the bottom of the graph.



Check: Dots sort themselves into groups, one dot per person!

One person = one row. Don't type the totals; CODAP counts for you.

MAKE

Objective 3 · Build step 2 of 2

Compare two groups

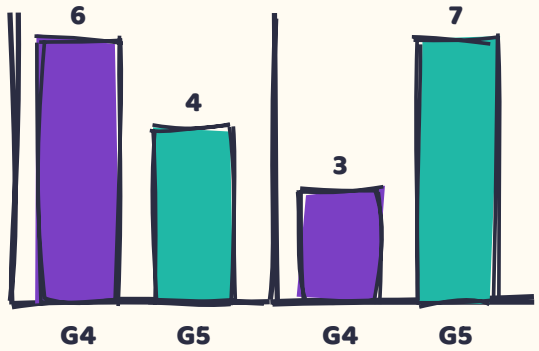
a Add your extra column (e.g. **grade** or **way to school**).

b Drag it onto the **side** (left axis) of your graph.

c Now you see the answers split into groups. Is the pattern the same in each group?

d Try a **number** question too (hours of sleep): CODAP can show the **mean** line (ruler menu).

e Take a screenshot of your best graph for your report.



Check: You can say something like 'Grade 5 likes football more than Grade 4 does'.

MAKE

Objective 4 · Conclude · report

Detective report

Report
Our question
How many people we asked
Our chart (screenshot / drawing)
Claim
Evidence (numbers!)
Limitation

Present it
Each team gets 1 minute. Show your chart and say your claim + evidence.

Test like an engineer

Detective's checklist

Check	✓ / ✗	Fix
Question is clear and fair		
Answer choices cover everyone		
Total = people asked		
Chart answers our question		
Claim uses numbers from the chart		
We named a limitation		

Debug checklist

CODAP graph shows one big blob? You dragged the 'name' column. Drag the answer column instead.

Privacy: use first names or initials only. Survey data about people should be kept kind and private.

MAKE

Level up!

Data challenges

Choose your challenge



Bigger sample: survey another class. Does the answer change?



Number data: hours of sleep: find the mean, the smallest and the biggest.



Two questions: do people who walk to school sleep more? Compare!



Infographic: make a poster with your chart and 3 key facts.

Think like an engineer

If only 5 people answered your survey, would you trust the result? Why not?

Boss level

Find a chart in a newspaper or online. Use the detective checklist on it.

Objective check

Can I really do it? Prove it.

01	Read out your survey question. Why is it fair?	○ ○ ○
02	Show your tally and table. Does the total match the class size?	○ ○ ○
03	Show your CODAP chart. What is the most common answer?	○ ○ ○
04	Say your claim + evidence, and one limitation.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Data gallery: every team's chart on the wall; others add a sticky note with one thing they learned.

Show what you know

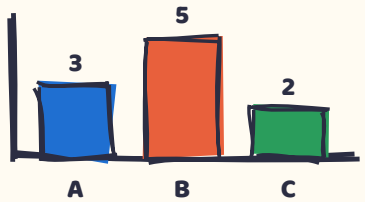
What did we learn?

- [x] Fair questions have clear choices and don't push an answer
- [x] Tally in 5s, then a table; check the total
- [x] Dot plots show every person; bar charts show totals
- [x] Claim + evidence (numbers!) + limitation
- [x] Check axes: charts can mislead

Exit ticket: a chart's axis starts at 50, not 0. Why might that be misleading?

At home

Survey your family on one question. Draw a bar chart and bring it in!



Next class: AI Emotion Mirror!